

Developer Mock Test Results

Student: Shiva kasula

Name:	Shiva kasula	Student ID:	112
Date:	2026-03-05 11:09:48	Performance:	■ Average
Overall Score:	10/25 (40.0%)	Test ID:	98799768-b865-45db...
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	2/10	20.0%	■■ Needs Work
■ MCQ/Theory	4/10	40.0%	■■ Needs Work
■ Coding	4/5	80.0%	■ Pass

Detailed Question Review

■ APTITUDE SECTION (2/10 - 20.0%)

■ Q1. A store buys a product for \$30 and sells it for \$25. What is the loss percentage?

Your Answer: 10%

Correct Answer: 16.67%

■ To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: $\text{Loss percentage} = ((\text{Cost price} - \text{Selling price}) / \text{Cost price}) * 100 = ((30 - 25) / 30) * 100 = (5 / 30) * 100 = 16.67\%$. The user's answer of 10% is likely incorrect because they may have made a calculation error or misunderstood the formula for loss percentage.

■ Q2. What is the next number: 1, 2, 4, 8, 16, ?

Your Answer: 36

Correct Answer: 32

■ The correct answer is 32 because the pattern of the sequence is obtained by doubling the previous term. The key calculation step is: $16 * 2 = 32$. The user's likely mistake is not recognizing the sequence as a powers-of-2 progression, instead possibly adding or multiplying by a different factor to arrive at 36.

■ Q3. If 8 workers can build a house in 12 days, how many days will it take for 6 workers to build the same house?

Your Answer: 14 days

Correct Answer: 16 days

■ To find the number of days it will take for 6 workers to build the house, we can use the formula: $(\text{number of workers})(\text{number of days}) = \text{constant}$. Given that 8 workers can build the house in 12 days, we have $(8)(12) = 96$. For 6 workers, let's denote the number of days as x , so $(6)(x) = 96$. Solving for x , we get $x = 96 / 6 = 16$. The user's mistake was

likely a calculation error, as the correct calculation yields 16 days, not 14.

■ **Q4. A mixture of paint and water is in the ratio 2:5. If 1 cup of paint is added to the mixture, what is the new ratio of paint to water?**

Your Answer: 3:5

■ **Correct!** Well done.

■ **Q5. A mixture is made of 3 parts coffee and 4 parts milk. What is the ratio of coffee to milk?**

Your Answer: 4:3

Correct Answer: 3:4

■ The correct answer is 3:4 because the mixture is made of 3 parts coffee and 4 parts milk, directly giving us the ratio of coffee to milk. The user's answer, 4:3, is likely a mistake due to reversing the given parts of coffee and milk. The key calculation step is simply comparing the given parts: coffee (3 parts) to milk (4 parts), resulting in a ratio of 3:4.

■ **Q6. A mixture of juice and water is in the ratio 3:5. If 3 cups of juice are added to the mixture, what is the new ratio of juice to water?**

Your Answer: 3:5

Correct Answer: 6:5

■ The correct answer is 6:5, which can be found by first representing the original ratio as $3x:5x$, where x is a multiplier. If 3 cups of juice are added, the new amount of juice is $3x + 3$, and since $3 = x$, the new amount of juice is $3x + x = 4x$, but to keep the ratio in simplest form and knowing the original ratio, we can calculate it as $(3x + 3)/(5x) = (3 + 3/x)/(5)$ and since $3/x$ is the original amount of juice over the ratio of juice to total parts ($3+5 = 8$), $3/x = 3/8 * 5/5 = 15/40 = 3/8$, so $3 + 3/x = 3 + 3/(3/5 * 8/8) = 3 + (5/3 * 8/8) = 3 + 5/3 = 14/3$, resulting in a ratio of $14/3 : 5 = 14 : 15$.

■ **Q7. If it is true that all cars are vehicles and all trucks are vehicles, what can be concluded about cars and trucks?**

Your Answer: They are different types of vehicles

Correct Answer: They are both vehicles

■ The correct answer, "They are both vehicles," is right because the given statements establish a common category between cars and trucks, which is that they are both vehicles. The user's answer, "They are different types of vehicles," is incorrect because while it may be true, it's not a conclusion that can be directly drawn from the given statements, which only establish their shared classification as vehicles. The user's mistake likely stems from adding an assumption about the differences between cars and trucks that isn't supported by the provided information.

■ **Q8. If a number is increased by 10% and then decreased by 10%, what is the overall percentage change?**

Your Answer: 1%

Correct Answer: -1%

■ The correct answer is -1% because when a number is increased by 10% and then decreased by 10%, the overall effect is a decrease. The key calculation step is: if we start with 100, increasing it by 10% gives 110, and then decreasing 110 by 10% gives 99, which is a 1% decrease from the original 100. The user's likely mistake was not considering the effect of the percentage decrease on the already increased value, rather than the original value.

■ **Q9. A shirt is on sale for 15% off its original price of \$80. How much will you pay for the shirt?**

Your Answer: \$65

Correct Answer: \$68

■ To find the sale price, calculate 15% of the original price of \$80: $0.15 * \$80 = \12 . Then, subtract the discount from the original price: $\$80 - \$12 = \$68$. The user's answer of \$65 is likely incorrect because they may have miscalculated the discount or subtracted an incorrect amount from the original price.

■ **Q10. A mixture is made of 2 parts water and 3 parts juice. What is the ratio of water to the total mixture?**

Your Answer: 2/5

■ **Correct!** Well done.

■ MCQ SECTION (4/10 - 40.0%)

■ Q1. What are the basic data types in Java?

Your Answer: int, double, char, Boolean

Correct Answer: int, double, char, String

■ The correct answer is int, double, char, String because these are the basic data types in Java that are commonly used to store whole numbers, decimal numbers, characters, and sequences of characters, respectively. The user's answer, int, double, char, Boolean, is incorrect because while Boolean is a data type in Java, it is not as fundamental as String in terms of basic data types. The user likely mistook Boolean for String, which is a more commonly used data type in Java for storing text.

■ Q2. What is the order of operations in Java?

Your Answer: Parentheses, multiplication and division, addition and subtraction, exponentiation

Correct Answer: Parentheses, exponentiation, multiplication and division, addition and subtraction

■ The correct answer is A) Parentheses, exponentiation, multiplication and division, addition and subtraction. This is because Java follows the standard order of operations, often remembered by the acronym PEMDAS (Parentheses, Exponents, Multiplication and Division, and Addition and Subtraction). The user's answer was incorrect, likely due to mistakenly placing exponentiation after multiplication and division, when in fact it should come before, as shown in the correct order.

■ Q3. What is an array in Java?

Your Answer: A collection of elements of different data types

Correct Answer: A collection of elements of the same data type

■ The correct answer is A) A collection of elements of the same data type, because in Java, an array is a data structure that stores a fixed-size collection of elements of the same data type. The user's answer, B) A collection of elements of different data types, is incorrect because Java arrays are strongly typed, meaning all elements in an array must be of the same data type. The user's likely mistake is a misunderstanding of Java's array type system, which does not allow for storing elements of different data types in the same array.

■ Q4. How should code be organized in Java?

Your Answer: Using methods and variables

Correct Answer: Using classes and packages

■ The correct answer, "Using classes and packages", is right because Java is an object-oriented language that emphasizes organization and structure through classes and packages. This allows for better code management, reusability, and readability. The user's answer, "Using methods and variables", is incorrect because while methods and variables are essential components of Java code, they are typically organized within classes and packages, rather than being the primary means of organization themselves.

■ Q5. What is exception handling in Java?

Your Answer: The process of handling syntax errors

Correct Answer: The process of handling runtime errors

■ The correct answer is A) The process of handling runtime errors, because exception handling in Java is a mechanism that allows developers to manage and respond to runtime errors, also known as exceptions, which occur during the execution of a program. The user's answer, B) The process of handling syntax errors, is incorrect because syntax errors are typically caught by the compiler before the program is even run, whereas exception handling deals with errors that occur at runtime. The user likely mistook exception handling for the compiler's ability to detect syntax errors, which are two distinct concepts in Java programming.

■ Q6. What is the base class for all exceptions in Java?

Your Answer: Error

Correct Answer: Throwable

■ The correct answer is C) Throwable, as it is the base class for all exceptions and errors in Java. The user's answer, B) Error, is incorrect because Error is a subclass of Throwable, but it is not the base class for all exceptions. The user likely mistook Error for the base class because it is a type of exception, but Throwable is the more general class that encompasses both exceptions and errors.

■ Q7. What are objects in Java?

Your Answer: Classes

■ *Well done!*

■ Q8. When should comments be written in Java?

Your Answer: For complex logic

■ *Correct! Well done.*

■ Q9. What is synchronization in Java?

Your Answer: The process of coordinating access to shared resources

■ *Excellent! Right answer.*

■ Q10. What should be followed in Java?

Your Answer: Java coding standards

■ *Correct! Well done.*

■ CODING SECTION (4/5 - 80.0%)

■ **Q1. Write a Java program that reads a string from stdin and checks if it's a palindrome. ****Input Format:**** - Line 1: A string ****Output Format:**** - "Yes"...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        String reversed = "";

        for (int i = input.length() - 1; i >= 0; i--) {
            reversed += input.charAt(i);
        }

        if (input.equals(reversed)) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }

        sc.close();
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and effective.*

■ **Q2. Write a program that reads a string of words and prints the frequency of each word. ****Input Format:**** - Line 1: Input string (string) ****Output Forma...****

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```

String input = sc.nextLine();
String[] words = input.split("\\s+"); // split by spaces

Map<String, Integer> freqMap = new LinkedHashMap<>();

for (String word : words) {
    freqMap.put(word, freqMap.getOrDefault(word, 0) + 1);
}

for (Map.Entry<String, Integer> entry : freqMap.entrySet()) {
    System.out.print(entry.getKey() + ":" + entry.getValue() + " ");
}

sc.close();
}
}

```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q3. Write a program that reads a student's name and their scores in 3 subjects, then prints their average score rounded to 2 decimal places. **Input Form...**

Your Answer:

```

import java.util.Scanner;
import java.text.DecimalFormat;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String name = sc.nextLine();
        int math = sc.nextInt();
        int science = sc.nextInt();
        int english = sc.nextInt();

        double average = (math + science + english) / 3.0;

        // Round to 2 decimal places
        DecimalFormat df = new DecimalFormat("0.00");
        System.out.println(df.format(average));

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q4. Write a Java program that reads a string from stdin and decrypts it using a simple substitution cipher. **Input Format:** - Line 1: A string **Output...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String encrypted = sc.nextLine();
        StringBuilder decrypted = new StringBuilder();
    }
}

```

```

        int shift = 3; // Caesar cipher shift

        for (char c : encrypted.toCharArray()) {
            if (c >= 'a' && c <= 'z') {
                char d = (char) (c - shift);
                if (d < 'a') d += 26; // wrap around
                decrypted.append(d);
            } else if (c >= 'A' && c <= 'Z') {
                char d = (char) (c - shift);
                if (d < 'A') d += 26; // wrap around
                decrypted.append(d);
            } else {
                decrypted.append(c); // non-alphabet characters remain unchanged
            }
        }

        System.out.println(decrypted.toString());
        sc.close();
    }
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String decrypted = decrypt(input);
        System.out.println(decrypted);
        scanner.close();
    }

    public static String decrypt(String input) {
        StringBuilder decrypted = new StringBuilder();
        for (char c : input.toCharArray()) {
            if (c == 'a') {
                decrypted.append('b');
            } else if (c == 'b') {
                decrypted.append('a');
            } else if (c == 'c') {
                decrypted.append('d');
            } else if (c == 'd') {
                decrypted.append('c');
            } else if (c == 'e') {
                decrypted.append('f');
            } else if (c == 'f') {
                decrypted.append('e');
            } else if (c == 'g') {
                decrypted.append('h');
            } else if (c == 'h') {
                decrypted.append('g');
            } else if (c == 'i') {
                decrypted.append('j');
            } else if (c == 'j') {
                decrypted.append('i');
            } else if (c == 'k') {
                decrypted.append('l');
            } else if (c == 'l') {
                decrypted.append('k');
            } else if (c == 'm') {
                decrypted.append('n');
            }
        }
    }
}

```

```

        } else if (c == 'n') {
            decrypted.append('m');
        } else if (c == 'o') {
            decrypted.append('p');
        } else if (c == 'p') {
            decrypted.append('o');
        } else if (c == 'q') {
            decrypted.append('r');
        } else if (c == 'r') {
            decrypted.append('q');
        } else if (c == 's') {
            decrypted.append('t');
        } else if (c == 't') {
            decrypted.append('s');
        } else if (c == 'u') {
            decrypted.append('v');
        } else if (c == 'v') {
            decrypted.append('u');
        } else if (c == 'w') {
            decrypted.append('x');
        } else if (c == 'x') {
            decrypted.append('w');
        } else if (c == 'y') {
            decrypted.append('z');
        } else if (c == 'z') {
            decrypted.append('y');
        } else {
            decrypted.append(c);
        }
    }
    return decrypted.toString();
}
}

```

■ The student's code is using a Caesar cipher with a shift of 3, but the expected output suggests a different shift or cipher. The actual output 'rovvy' is 13 positions ahead of the input 'uryyb' in the alphabet, indicating a possible shift of 13 instead of 3. This discrepancy is causing the output to differ from the expected 'hello'.

■ **Q5. Write a Java program that reads a string from stdin and counts the frequency of each word, using a lambda expression to sort the results. **Input For...**

Your Answer:

```

import java.util.*;
import java.util.stream.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();
        String[] words = input.split("\\s+");

        // LinkedHashMap preserves insertion order
        Map<String, Integer> freqMap = new LinkedHashMap<>();
        for (String word : words) {
            freqMap.put(word, freqMap.getOrDefault(word, 0) + 1);
        }

        // Sort by frequency descending, preserve order for equal frequency
        freqMap.entrySet().stream()
            .sorted((a, b) -> b.getValue().compareTo(a.getValue()))
            .forEach(entry -> System.out.print(entry.getKey() + ":" + entry.getValue() + " "));
    }
}

```

```
        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an overall "Accepted" status, keep up the good work!

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
 - **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
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